

# Probability of Winning a Game of “Flip”

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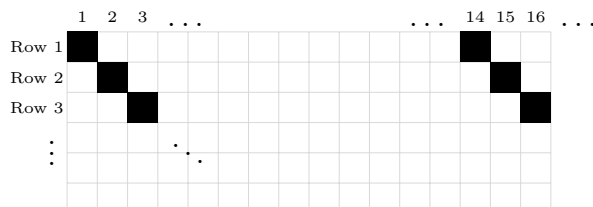
## 1 Problem Overview

The game Flip proceeds as follows:

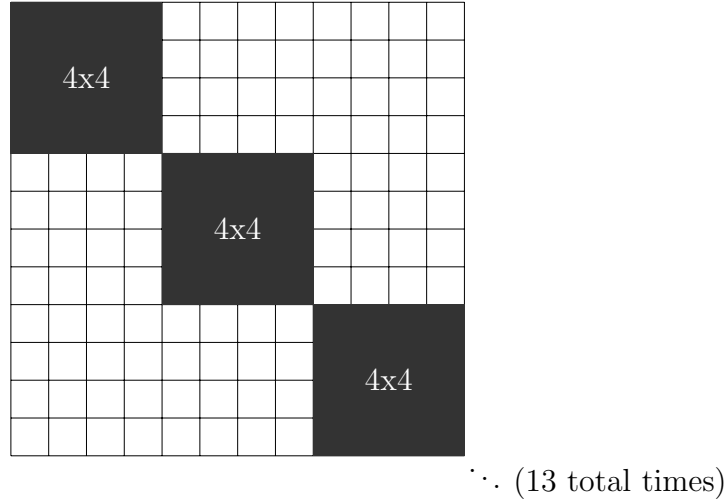
1. A standard deck is shuffled.
2. The player names ranks in a fixed sequence: *Ace, 2, 3, ..., King, Ace, 2, ...* until the deck is exhausted.
3. On each turn, the player flips the top card. If the rank named matches the rank of the flipped card, the player “dies.”
4. The goal is to survive all 52 turns.

## 2 Chess Boards

To calculate the probability of a successful run-through, we set up a  $52 \times 52$  chess board for rook polynomials. There are 4 squares that are blacked out on each row and each column, representing the 4 spots that a card cannot be mapped to (*see below for my best attempt at rendering this into LaTeX*).



We can rearrange the board so that we have 13  $4 \times 4$  grids of blacked out squares that don’t share any rows or columns:



### 3 The Rook Polynomial

To place  $n$  rooks on a blacked out square, we take the coefficient of  $x^n$  in the following expression:

$$(1 + 16x + 72x^2 + 96x^3 + 24x^4)^{13}$$

The term within the parenthesis is the rook polynomial of a  $4 \times 4$  grid. Since the  $4 \times 4$  grids don't share rows and columns, we can multiply their rook polynomials (hence the exponent of 13) to find  $r(\bar{C})$ . Let the coefficient of  $x^n$  in the final expanded polynomial be  $c(n)$ .

### 4 Inclusion-Exclusion and Probability

Now we use the Principle of Inclusion-Exclusion (PIE) to calculate the number of "successes" where no card is deranged to its same location (e.g., a 2 to any of the 4 "2" spots). The number of successes is:

$$52! - 51! \cdot c(1) + 50! \cdot c(2) - \dots + c(52)$$

Then we can take the total number of deck arrangements ( $52!$ ) to calculate the probability of success using a constant distribution:

$$P(\text{Success}) = \frac{\sum_{n=0}^{52} (-1)^n \cdot (52 - n)! \cdot c(n)}{52!}$$